

Trump's posts and the price of crude oil

Project status

updated on 05/10/2026

1 The processing chain

The question is whether Trump's posts that signal a risk to crude supply move the price of oil, and for how long. Three things are needed: knowing *when* each post was published, precisely enough to isolate its effect; establishing *which* posts count as events, with a replicable criterion; having a price series dense enough to measure the reaction in the following minutes.

The work is split into numbered scripts, each reading the output of the previous one.

00 corpus

Downloads the posts from the archive of the American Presidency Project (UCSB). The publication time is not read from the page but derived from the post identifier, which contains it by construction. The time printed on the page is only a check.

00b integration

Adds the posts UCSB does not have, taken from CNN's public archive. Here too the time comes from the identifier, extracted from the post link: the two sources are treated the same way.

01 selection

Discards what has no text content, applies the rule that defines an oil event, groups close posts into episodes and measures how long the observation windows can be. The episode is the unit of analysis.

02 audit

Tests the rule with two unsupervised methods that do not know the term lists. It measures how much the rule misses; it does not define anything.

03 prices

Downloads the one-minute price series. On hold, pending a decision on which series to use.

04 features

Measures each episode: time, length, emphasis, novelty with respect to recent episodes, and the variables that require reading (expected direction on supply, intensity, specificity).

Two support folders: **annotations/** keeps every version of the rule together with the annotated sample used to evaluate it; **query/** holds read-only scripts to query the data.

The features must be frozen before looking at prices: tuning them on returns would mean building the result instead of measuring it.

2 Where we are

Step	Status	Output
00–00b corpus	done	13,014 posts up to 01/10/2026
01 selection	done, rule v6	368 events, 323 episodes
02 audit	done	49 clusters, 95 candidate false negatives

03 prices	on hold	test on March 2025
04 features	steps A, B, C	20 columns, gold standard of 145 episodes

3 The corpus

Sources

Source	Posts
American Presidency Project (UCSB)	8,619
CNN archive, only posts missing from UCSB	4,395
Total, from 1/12/2024 to 01/10/2026	13,014

UCSB is a curated, incomplete archive. Between December 2024 and July 2026 the CNN archive contains 2,811 posts that UCSB does not have; UCSB therefore covered about 75% of the actual output. Posts after 31 July 2026 come on top of these. Truth Social blocks automated access to its API (403 response), so it is not used as a direct source.

The timestamp

Truth Social uses snowflake identifiers, whose high bits are the milliseconds since the Unix epoch:

$$t_{UTC} = \frac{\text{post_id} \gg 16}{1000}, \quad \text{seq} = \text{post_id} \& 0\text{xFFFF}.$$

The sequence field orders posts published within the same millisecond. For both sources the identifier is extracted from the post link.

Checks: the announcement of the strike on Fordow comes out at 19:50 EDT on 21 June 2025, the publicly known time; the date declared by CNN matches the one derived from the ID for every post, within one second; for UCSB the time on the page matches in 98.1% of cases.

UCSB anomalies

The archive prints the time in two formats: 24-hour (**18:11**) and 12-hour with a meridiem (**1:17 PM**). The first version of the parser ignored the meridiem, and the suffix ended up at the start of the post text. Fixed. The remaining anomalies do not affect the timestamps, which come from the ID:

Offset	Rows	Cause
180 min	105	UCSB prints Eastern time from 15 to 21 September 2025
other	36	isolated editorial errors
missing	15	no time on the page
720 min	8	24-hour page with a wrong time (12:27 for 00:27)

4 Cleaning

Of the 13,014 posts, 5,950 remain (45.7%).

Criterion	Removed	What it removes
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No content	4,311	videos and images without a caption, placeholders
Double posts within 2 minutes	563	identical texts close in time
Endorsement boilerplate	975	the template of election endorsements
Under 5 words	1,215	slogans and fragments

Most removals are posts without text: Trump publishes many videos and images without writing anything. Once the file address is removed, an empty string or the word **Video** is left. Excluding them is unavoidable for a lexical rule, but it is a limit to declare: a video of a refinery being hit is market information, and no word-based rule can see it.

Open problems

Double posts. The filter compares normalized texts (lowercase, no punctuation). All video-only posts have the same normalized text, **video**, and two different videos less than two minutes apart are merged as if they were a double post. The result does not change, because those posts would be discarded anyway, but it makes the counts in the table hard to read.

Reposts. The corpus contains 752 reposts (RT): 460 of Trump’s own posts, 291 of other accounts, the rest links only. Reposts of his own posts are duplicates: same text, different identifier, published minutes or hours after the original. Today they enter as distinct events (see novelty, section 9). In the CNN format, moreover, the account name is glued to the text (RT @realDonaldTrumpIran is...), and removing mentions deletes the first word of the post.

5 The rule

A post is an event if it contains at least one supply actor and at least one mechanism:

$$\text{event}_i = \mathbb{1}[|A \cap T_i| \geq 1 \wedge |M \cap T_i| \geq 1],$$

with T_i the tokens of the post, A the list of actors, M the list of mechanisms.

- **Actors:** Iran and Hormuz, Venezuela, Russia and Ukraine, OPEC and Saudi Arabia, the strategic reserve, routes (Red Sea, Houthis, Suez), Nigeria, explicit oil (**oil**, **crude**, **barrel**, **gasoline**, **fuel**).
- **Mechanisms:** physical disruption (blockade, mines, tankers, strait, ports, pipelines), military action, sanctions and tariffs, nuclear negotiation, regime change, ultimatum, flows resuming.

On the clean corpus: 782 posts with an actor, 892 with a mechanism, **368 events**. Of these, 141 rest only on weak mechanisms (**military**, **attack**, **strike**, **nuclear**): conflict terms that do not directly concern supply. Even so, most of these posts are of real interest, so weak mechanisms remain a valid tool.

Annotation criterion

A post is relevant (label 1) if it concerns:

1. **the conflict with Iran and the Strait of Hormuz:** attacks, damage, ceasefires, negotiations, opening or closing of the strait. They shape expectations on the most important oil route.
2. **the Iranian nuclear programme:** including statements such as “Iran will never have a nuclear weapon”, read as a sign of tension between Iran and the US.
3. **Venezuela:** the action of the Trump administration has affected Venezuelan oil flows.

4. **Russia and Ukraine**, when they touch sanctions, energy or the end of the war.

It is not relevant (label 0):

1. an actor or mechanism mentioned in passing in a post about something else (Bondi, Epstein, the Kennedy Center);
2. polls and linked op-eds, arguments with the media and opponents about facts already known;
3. military facts unrelated to supply (strikes on drug-trafficking boats, Russian drones over Poland);
4. comments on prices that have already moved (“Oil prices are down”): the information flows from the price to the post, not the other way round;
5. posts about Greenland, for now.

Versions

Each version saves in **annotations/** the rule and the annotated sample: 100 posts captured by the rule (*inside*) and 50 meeting only one criterion (*boundary*). The share of relevant posts *inside* is the precision; the share on the *boundary* estimates how many relevant posts the rule misses. 95% confidence intervals (Wilson).

	Added	Removed	Precision	Relevant on the boundary
v1	—	—	85% 77%–91%	8% 3%–19%
v2	droppingregime change, obliterate, obliterated, obliterating, obliteration	dropping	80% 71%–87%	10% 4%–21%
v3	clock is ticking, deadline, regime change, to flow, ultimatum	boom, down, droppingregime change, market, markets, plummet	74% 65%–82%	14% 7%–26%
v4	nigeria, tariff	—	80% 71%–87%	12% 6%–24%
v5	—	—	81% 72%–88%	16% 8%–28%
v6	—	mine	85% 77%–91%	8% 3%–19%

With the current version the *boundary* counts 938 posts. The estimated recall is about **81%**: an upper bound, because it does not count relevant posts with neither actors nor mechanisms. With 100 and 50 annotated posts the intervals are wide, and the differences between neighbouring versions are not significant.

Structure of the rule

The rule errs more by excess than by omission: Trump also uses the keywords to attack political opponents. On 482 posts annotated across all versions, proximity constraints between actor and mechanism (same sentence, within 20 words) remove almost as many relevant posts as irrelevant ones. False positives are not a lexical problem: in “Strong

Majority Backs Trump in Stopping Iran’s Nuclear Ambitions” actor and mechanism are adjacent, and it is the purpose of the post that makes it irrelevant.

A semantic classifier (logistic regression on embeddings) trained on the same 482 labels separates relevant and irrelevant posts with an AUC of 0.92 in cross-validation. Used as a second stage, it raises precision *inside* from 80% to 87% while losing 6 relevant posts out of 217. It is an open decision (section 10).

6 Unsupervised audit

Last run on the full clean corpus (5,950 posts): 49 clusters, 34.9% outliers. The rule covers 60% of the largest Iran/Hormuz cluster (226 of 379 posts).

Clusters with an oil lexicon and low coverage:

Characteristic terms	Captured	Coverage
ukraine, russia, putin, war, president putin, zelenskyy	18 / 77	23%
venezuela, oil, strike, united states, united, states	16 / 69	23%
epa, energy, zeldin, lee zeldin, lee, administrator lee	0 / 57	0%
prices, gas, gas prices, gasoline, walmart, oil	1 / 29	3%

Russia/Ukraine and Venezuela are inside the scope of the thesis: their low coverage is a real gap in the term lists, consistent with the posts on the Russia–Ukraine peace process found on the *boundary*, which name the actors but no mechanism. The energy-policy cluster (EPA) and the cluster on pump prices are outside the scope by construction.

The semantic search returns 95 candidate false negatives, posts close to the probe sentences but not captured by the rule.

7 Episodes, window, power

Posts within 30 minutes form one episode, dated at the first post: 368 events give **323 episodes**, 41 of which with more than one post. Clustering in time is marked (June 2025, March–June 2026) and provides the regimes to compare with CD-NOD.

Episodes preceded by another within one hour: 5.6% within two hours: 13.7%. With the 15% threshold the event window reaches two hours.

Minimum detectable effect, two-sided test, $\alpha = 0.05$, power 0.8:

$$\text{MDE} = \frac{z_{1-\alpha/2} + z_{1-\beta}}{\sqrt{n}} \sigma = \frac{2.802}{\sqrt{n}} \sigma.$$

With $n = 323$: 0.156 standard deviations. On episodes with a WTI price (68.1%): 0.189.

8 Price data

On hold. Test on March 2025, files in `download/`. A bar counts as real if it is not inside a constant-price stretch of at least one hour: Dukascopy fills closed-market periods by repeating the last price.

- **Brent**: unusable. Every Tuesday and Thursday of the month is missing, and the days present stop at 21 UTC.
- **WTI**: covers from Sunday 22 UTC to Friday 21, with a daily break between 21 and 22. The Sunday bars before 22 are filler.
- **Gold**: same structure, downloaded for one week only.

68.1% of episodes fall at hours when WTI has a price (`query/post_coverage.py`). The `us_market_open` variable (30.3% of episodes) measures the equity session, not the oil one, and does not indicate how many episodes are lost.

Open questions:

1. **WTI coverage** beyond March 2025, not verified.
2. **Nature of the data:** these are a broker’s CFD quotes, not exchange prices. Over windows of minutes the difference should be negligible, but it needs to be endorsed.
3. **Episodes without a price:** about a third. Accept the loss, look for a more continuous instrument, or change the design for that slot?

9 Features

`episode_features.parquet`: 323 episodes, 20 columns. Steps A (deterministic features), B (novelty) and C (gold standard sample) are computed.

Novelty

$$\text{novelty}_i = 1 - \max_{j: t_i - 30 \text{ days} \leq t_j < t_i} \cos(e_i, e_j),$$

with e_i the embedding of the episode text. Median 0.381, 10th percentile 0.175, 90th 0.637. The values are relative: the variable serves as a ranking and has no natural threshold. Near 0 is the same text republished; around the median the same topic with different content; above 0.6 content with no recent precedent.

30 episodes have novelty below 0.15, and 22 of them are reposts of Trump’s own posts: duplicates, not repetitions of content. The link with text length is weak (Spearman -0.11), but length stays as a control.

Gold standard

145 episodes exported instead of 200: the quota per quarter is fixed and quarters with few episodes do not fill it. Annotation follows `annotations/gold_guidelines.md`. Before annotating, reposts of his own posts should be removed.

10 Decisions to take together

1. **The price series** (section 8). Until it is decided, the dependent variable does not exist.
2. **Rule or rule plus classifier.** The semantic second stage improves precision, but the definition of event becomes what the model learned from 482 examples, and it must be frozen and declared like a prompt.
3. **The boundary of the topic.** Pump prices, market comments, Greenland: today they are excluded by the annotation criterion. To be confirmed.
4. **The Eastern-time week.** Declared exception in the validator or documented anomaly.

11 Planned work

1. Remove reposts of his own posts during cleaning; add a column distinguishing reposts of other accounts.
2. Make the double-post filter conditional on the presence of text.

3. Extend the term lists for Russia/Ukraine and Venezuela, the two gaps found by the audit.
4. Fix the gold standard export to reach 200 episodes; annotate it, run the LLM scoring, measure the weighted kappa.
5. Freeze the features and record the date.

12 Change log

5/10/2026

Repository translated into English: files, folders, data columns and the values `dentro/confine` (now `inside/boundary`). Data contents unchanged. Old-to-new names in `GLOSSARY.md`.

3/10/2026

Audit rerun on the full corpus. Annotation guidelines for the gold standard (`annotations/gold_guidelines.md`).

2/10/2026

Rule v6: removed `mine`, which also fired as a pronoun (“not mine”). Semantic novelty and gold standard sample computed. Scripts `query/series_hours.py` and `query/post_coverage.py` on the hourly coverage of the series.

1/10/2026

4,395 posts integrated from the CNN archive (`00b`), with the timestamp from the link. Rule v4: added `nigeria` and `tariff`. Rule v5: same rule, sample regenerated on the integrated corpus.

29/9/2026

Rule v3: removed `market`, `markets` and the price terms (`down`, `boom`, `plummet`, `dropping`); they comment on prices that have already moved. Added `regime change`, `ultimatum`, `deadline`, `clock is ticking`, `to flow`. In v2 a missing comma had merged `dropping` and `regime change` into a single term (`droppingregime change` in the versions table): neither was active.

29/9/2026

Rule v2: added `obliterate` and its variants.

25/9/2026

Created the `annotations/` folder: each version keeps the rule and the annotated sample; `fill_labels.py` carries labels already given over to new samples. Rule v1 annotated, already without `war`: too generic, it fired on any war.

21/9/2026

First draft. Fixed the parsing of 12-hour times in `00`; recovered 12 posts lost to the timestamp format in `01`.